

Effects of lanthanide ions on the structure and electrical properties of Aurivillius $\text{Bi}_3\text{TiNbO}_9$ high temperature piezoelectric ceramics

Dingwei Hou, Huiqing Fan*, Yanqin Chen, Yuxin Jia, Weijia Wang*

State Key Laboratory of Solidification Processing, School of Materials Science and Engineering, Northwestern Polytechnical University, Xi'an 710072, China

ARTICLE INFO

Article history:

Received 16 April 2022

Received in revised form 6 June 2022

Accepted 26 June 2022

Available online 2 July 2022

Keywords:

Bismuth layer structured ferroelectrics

Rietveld refinement

Lanthanide ions

Band structure

ABSTRACT

Aurivillius $\text{Bi}_3\text{TiNbO}_9$ has a very high Curie temperature (T_c), but the piezoelectric coefficient (d_{33}) is very low, and the influence of ions with different radii on its performance is not clear. High temperature piezoelectric ceramics $\text{Bi}_{2.99}(\text{Li}_{0.5}\text{Ln}_{0.5})_{0.01}\text{TiNbO}_9$ (BTNO-1LiLn, Ln = La, Nd, Gd, and Er) were prepared by solid state reaction sintering, and its microstructure, piezoelectric properties and electrical conductivity were systematically studied. The T_c of BTNO-1LiLn increases with the decrease of Ln^{3+} ions radii. BTNO-1LiEr obtains the highest d_{33} of 10.6 pC/N. Considering the cation disorder of A-site and bismuth layer of calcium titanium layer in Aurivillius structure, the crystal structures of BTNO-1LiLa with special occupation and BTNO-1LiEr with the best performance are refined, and the source of spontaneous polarization (P_s) is analyzed. It is proved that different occupation will have different effects on the high temperature piezoelectric properties and fatigue cycle properties of BTNO based ceramics. Finally, the band structure of BTNO pure ceramic based on the first principle is calculated. By comparing the activation energy and band gap width, it is indicated that $(\text{Li}_{0.5}\text{Ln}_{0.5})^{2+}$ ions entering BTNO will form impurity energy levels and form p -type semiconductors. This study provides a theoretical basis for better performance of BTNO based high temperature piezoelectric ceramics.

© 2022 Elsevier B.V. All rights reserved.

1. Introduction

Piezoelectric ceramic materials are a kind of important functional materials to realize electromechanical energy conversion and coupling. Using the positive piezoelectric effect of mechanical vibration converted into electrical signal, it can be used to manufacture piezoelectric pickup, buzzer, and ultrasonic receiving probe [1–4]. Conversely, by converting the AC signal into the inverse piezoelectric effect of mechanical vibration, it can be used to manufacture ultrasonic emitters, piezoelectric speakers, and ultrasonic cleaning agents [5–8].

In the current practical application, lead zirconate titanate (PZT) occupies a dominant position, but its curie temperature (T_c) is generally maintained at about 350 °C, so the temperature of PZT working stably for a long time is also around 170 °C, and serious depolarization will occur when the ambient temperature is higher than 170 °C. Moreover, lead is easy to volatilize at high temperature, which will also pollute the environment and cause damage to human body.

Therefore, lead-free piezoelectric materials that can work normally in a wider temperature range have become a research hotspot. The general structural formula of bismuth layer structured ferroelectrics (BLSFs) discovered by Aurivillius is $(\text{Bi}_2\text{O}_2)^{2+}(\text{A}_{n-1}\text{B}_n\text{O}_{3n+1})^{2-}$ [9], the A-site is the cubic octahedral coordinated +1, +2, and +3 valence ions or their combined ions, the B-site is the transition group ions or their combined ions suitable for octahedral coordination, and the bismuth layer of $(\text{Bi}_2\text{O}_2)^{2+}$ and the calcium titanium layer of $(\text{A}_{n-1}\text{B}_n\text{O}_{3n+1})^{2-}$ are regularly arranged alternately, n is the number of combined layers [10,11]. BLSFs has high dielectric breakdown strength, high T_c and low aging rate, which makes BLSFs could work continuously at high temperature as piezoelectric material [6]. Xiangxiong Zeng et al. [12] prepared a series of ceramics $\text{Ca}_{0.85}(\text{Li}_{0.5}\text{Ce}_{0.25}\text{Nd}_{0.25})_{0.15}\text{Bi}_2\text{Nb}_2\text{O}_9$, piezoelectric coefficient (d_{33}) was increased from 4 pC/N to 13.1 pC/N, but T_c was reduced from 933 °C to 900 °C. Zhihang Peng et al. [13] introduced $(\text{Li}_{0.5}\text{Ce}_{0.25}\text{Pr}_{0.25})^{2+}$ combined ions into $\text{CaBiNb}_2\text{O}_9$, they found that when $x = 0.10$, d_{33} reaches 17.3 pC/N and T_c rises to 938 °C. It can be seen that good properties of BLSFs can be obtained by selecting the smallest alkali metal ion Li^+ and rare earth ions to form complex-ions for modification. However, the effect of different rare earth ions on the high temperature piezoelectric properties of BLSFs at the same doping concentration is not clear, although they have similar electronic structures. $\text{Bi}_3\text{TiNbO}_9$ ($n = 2$, abbreviated as BTNO) is

* Corresponding authors.

E-mail addresses: hqfan@nwpu.edu.cn, hqfan3@163.com (H. Fan), weijia.wang@nwpu.edu.cn (W. Wang).

one of the ceramics with T_c above 900 °C in BLSFs, but its piezoelectric coefficient (d_{33}) is low (~ 6.0 pC/N), which limits its further use [14]. This is due to its limited spontaneous polarization along the a -axis. The d_{33} of BLSFs ceramics prepared by grain orientation technology, such as template grain growth method (TGG) and spark plasma sintering (SPS), can be significantly improved, but its high conductivity along the maximum d_{33} direction makes the polarization process difficult, and the process is more complex and cost-effective [15–18]. Therefore, it is more desirable to obtain BLSFs ceramics from the traditional solid state reaction sintering.

To sum up, according to our previous work [19], when the coordination number is 12, La^{3+} , Nd^{3+} , Gd^{3+} , and Er^{3+} with different radii were selected to form $(\text{Li}_{0.5}\text{Ln}_{0.5})^{2+}$ combined ions with Li^+ respectively, and replace Bi^{3+} at A-site of $\text{Bi}_3\text{TiNbO}_9$ with the same concentration ($x = 0.01$) to obtain a series of ceramics $\text{Bi}_{2.99}(\text{Li}_{0.5}\text{Ln}_{0.5})_{0.01}\text{TiNbO}_9$ ($\text{Ln} = \text{La, Nd, Gd, and Er}$). The effect of $(\text{Li}_{0.5}\text{Ln}_{0.5})^{2+}$ on the high temperature piezoelectric properties of $\text{Bi}_3\text{TiNbO}_9$ ceramics was clarified, and the structure and conductive mechanism of BTNO-1LiLn ceramics were studied.

2. Experiment

$\text{Bi}_{2.99}(\text{Li}_{0.5}\text{Ln}_{0.5})_{0.01}\text{TiNbO}_9$ (BTNO-1LiLn, $\text{Ln} = \text{La, Nd, Gd, and Er}$) high temperature piezoelectric ceramics were prepared by traditional solid state reaction sintering method. According to the stoichiometric ratio, the high purity starting materials Bi_2O_3 (99.5 %, Sinopharm Chemical Reagent Co., Ltd.), TiO_2 (99.0 %, Sinopharm Chemical Reagent Co., Ltd.), Nb_2O_5 (99.5 %, Sinopharm Chemical Reagent Co., Ltd.), Li_2CO_3 (99.9 %, Sinopharm Chemical Reagent Co., Ltd.), and La_2O_3 (99.9 %) [or Nd_2O_3 (99.9 %), Gd_2O_3 (99.9 %), Er_2O_3 (99.9 %), Sinopharm Chemical Reagent Co., Ltd.] were mixed and milled in ethanol for 12 h. The obtained slurry was dried, screened, and calcined at 900 °C for 6 h. After the second ball milling in ethanol for 12 h, the pulp obtained for the second time was dried and screened, and then pressed into a disc with a diameter of 12 mm and a thickness of 1 mm. These round pieces are pressed by a cold isostatic press at a pressure of 200 MPa. Finally, the obtained powder was sintered for 3 h in the temperature range of 1115 °C $\sim$ 1125 °C.

Archimedes method was applied to measure densities of the sintered ceramics using ethanol as a medium, and then their relative densities were calculated. The crystal structure of ceramics at the temperature of 25 °C was characterized by X-ray diffractometer (XRD; X'pert PRO MPD, Philips, Eindhoven, The Netherlands). The test range is 20–80° and the test speed is 3°/min. The lattice vibration properties of ceramic samples at the temperature of 25 °C were studied by Raman spectroscopy (LabRAM, Horiba Jobin-Yvon, Lyon, France). The microstructure of ceramics can be observed by field emission scanning electron microscope (FE-SEM; Tescan Clara GMH, Czech Republic). The photoelectron spectra of the elements were detected by X-ray photoelectron spectroscopy (K-alpha, Thermo Scientific, England). The sintered ceramics were ground to 0.4 mm and polished, then they were painted with silver paste and cured at a temperature of 500 °C for 30 min to form the electrode. Finally, the samples coated with silver electrode were poled under the DC electric field of 100 kV/cm at 180 °C for 20 min. The piezoelectric coefficient (d_{33}) was measured by quasi-static d_{33} meter (ZJ-4A, Institute of Acoustics, Chinese Academic of Science, Beijing, China). And Curie temperature (T_c) of ceramic samples is obtained from the dielectric temperature spectrum. The dielectric permittivity (ϵ') and loss tangent ($\tan \delta$) of ceramic samples were measured by impedance analyzer in the temperature range of 25 °C $\sim$ 935 °C at 100 kHz, which was achieved by heating in muffle furnace. The fatigue cycle properties of series ceramics were tested by TF-2000 (Aix ACCT, Aachen, Germany). The resonant and anti-resonant frequencies of the ceramics were measured by Agilent 4294 A precision impedance analyzer, and then the plane electromechanical coupling

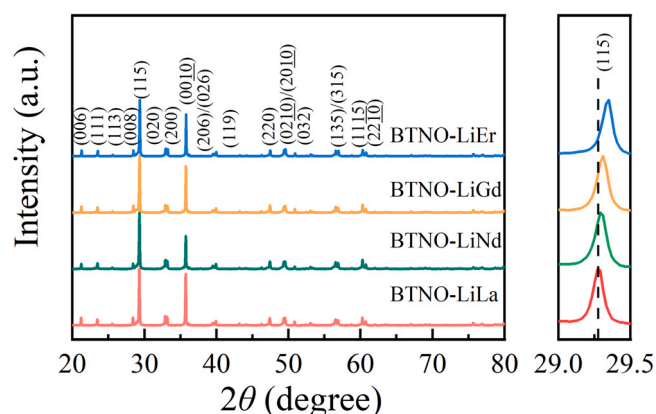


Fig. 1. XRD patterns of BTNO-1LiLn ($\text{Ln} = \text{La, Nd, Gd, and Er}$) ceramics at 25 °C in the range of (a) 20° $\sim$ 80°, and (b) 29.0° $\sim$ 29.5°.

factor (k_p) and thickness electromechanical coupling factor (k_t) of ceramics were calculated. The conductive mechanism of ceramic samples can be analyzed by complex impedance spectrum, which could be tested by AC impedance analyzer (SI 1260, Solartron, Hampshire, UK), and frequency ranges from 0.1 Hz to 1 MHz.

3. Results and discussion

3.1. Phase, microstructure and morphology analysis

Fig. 1(a) shows XRD pattern of $\text{Bi}_{2.99}(\text{Li}_{0.5}\text{Ln}_{0.5})_{0.01}\text{TiNbO}_9$ ($\text{Ln} = \text{La, Nd, Gd, and Er}$) ceramics. As shown in the pattern that all diffraction peaks can be indexed to the orthorhombic structure of $\text{Bi}_3\text{TiNbO}_9$ ($A2_1am$; ICSD # 79–1550). As shown in Fig. 1(b), when the major peak (115) from 29° to 29.5° is enlarged, it is clear that there is no peak splitting and that the peak shifts to high angles as the atomic number increases. Since the ion radii of Gd^{3+} and Er^{3+} at CN = 12 have not been reported, linear fitting is carried out according to their ion radii at other coordination numbers [20], as shown in Fig. 2(a) and (b). According to Bragg equation $2d\sin \theta = n\lambda$, it indicates that $(\text{Li}_{0.5}\text{La}_{0.5})^{2+}$ (1.26 Å, CN = 12), $(\text{Li}_{0.5}\text{Nd}_{0.5})^{2+}$ (1.215 Å, CN = 12), $(\text{Li}_{0.5}\text{Gd}_{0.5})^{2+}$ (1.192 Å, CN = 12), and $(\text{Li}_{0.5}\text{Er}_{0.5})^{2+}$ (1.1695 Å, CN = 12) have successfully substituted Bi^{3+} (1.38 Å, CN = 12), respectively.

The Raman spectra of BTNO-1LiLn ceramics at 25 °C in the wave number range of 0 cm^{-1} $\sim$ 1000 cm^{-1} is shown in Fig. 3. Table 1 shows the different mode of vibrations for BTNO based ceramics, the soft mode of phase structure transformation is below 30 cm^{-1} , the phonon of 62 cm^{-1} comes from the vibration of Bi^{3+} ions in bismuth layer, and 80 cm^{-1} $\sim$ 160 cm^{-1} corresponds to the vibration of A-site ions; 230 cm^{-1} corresponds to the twisting mode of BO6, 529 cm^{-1} $\sim$ 563 cm^{-1} corresponds to the stretching mode of xy diagonal oxygen ions of BO6 octahedron, and 833 cm^{-1} corresponds to the tensile mode of the BO6 octahedron [21]. All ceramics show similar phonon vibration, which indicates that the substitution of Bi^{3+} by low concentration $(\text{Li}_{0.5}\text{Ln}_{0.5})^{2+}$ ($\text{Ln} = \text{La, Nd, Gd, and Er}$) ions does not significantly change the vibration mode of bond in BTNO based ceramics. When Lorentz fitting is applied to peak positions in the range of 150 cm^{-1} $\sim$ 350 cm^{-1} , it can be seen from Fig. 3(b) that the peak has a very slight red shift, which is due to the shorter bond length with the increase of the atomic number of the dopant, so it moves to the higher wave number [22,23]. Both indicate that $(\text{Li}_{0.5}\text{Ln}_{0.5})^{2+}$ ions are successfully dissolved into BTNO.

Fig. 4 shows the SEM of BTNO-1LiLn ceramics. It can be observed from the figure that the grain growth shows very obvious disorder. The grain morphology is bismuth layered structure and typical plate layered structure of ferroelectrics. This is because its growth rate along the a - b plane is much higher than that along the c axis.

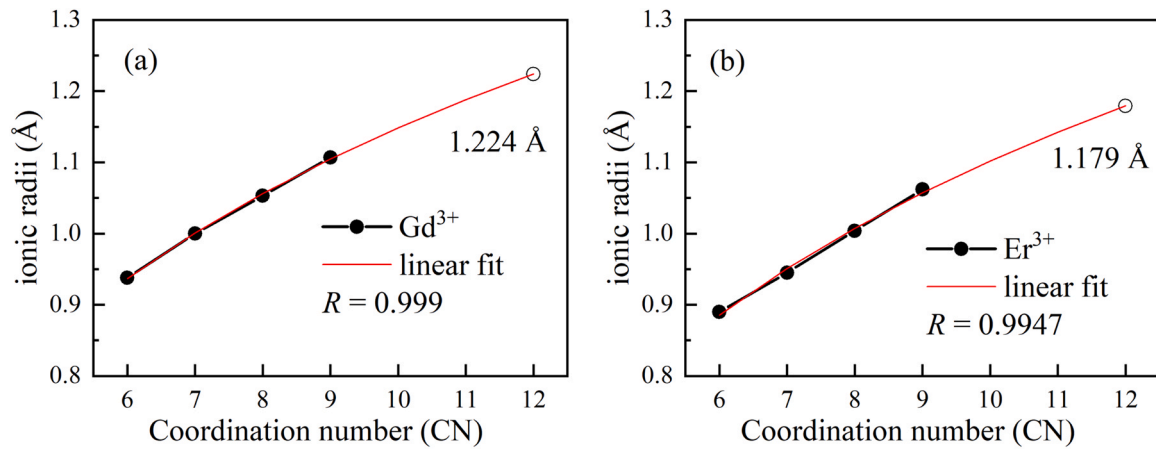


Fig. 2. The linear fitting of ion radii under different coordination numbers of (a) Gd^{3+} and (b) Er^{3+} .

Because the lamellar grains with various orientations are stacked together, it is difficult to quantitatively give the distribution of grain size from SEM. From the SEM images after thermal etching, it can be seen that all samples have a certain degree of holes at the grain boundary, which is due to the melted surface is polished, exposing the internal gaps caused by the stacking of plate-like grains.

3.2. Piezoelectric properties

The dielectric permittivity (ϵ') and loss tangent ($\tan \delta$) of BTNO-1LiLn ceramics at 25 °C to 935 °C are shown in Fig. 5. As can be seen from the inserted in Fig. 5(a), that the curie temperature (T_c) of ceramics doped with different rare earth elements is well maintained above 915 °C and the ϵ' are also maintained above 1000. This is due to the lanthanides have similar properties [24,25]. Fig. 5(b) shows that the loss peak for all ceramics appears around T_c , and the inserted shows that the loss remains low until 575 °C, indicating extremely strong high temperature performance.

Fig. 6(a) is the effect of annealing temperature on the piezoelectric coefficient (d_{33}) of BTNO-1LiLn. After annealing at different temperatures for 2 h and measuring their d_{33} at room temperature, BTNO-1LiEr showed the maximum d_{33} . As can be seen from the normalized d_{33} in Fig. 6(b), all ceramics show very good temperature

Table 1

Different mode of vibrations for BTNO based ceramics.

Wavenumber (cm^{-1})	Mode of vibrations
0 ~ 30	The soft mode of phase structure transformation.
62	The vibration of Bi^{3+} ions in bismuth layer.
84, 110, and 183	The vibration of A-site ions.
230	The twisting mode of BO6.
529 ~ 563	The stretching mode of xy diagonal oxygen ions of BO6 octahedron.
833	The tensile mode of the BO6 octahedron.

stability and remain well until 800 °C. When the temperature rises higher, depolarization occurs, which is manifested as the decrease of d_{33} . Among them, the relative d_{33} of BTNO-1LiNd, BTNO-1LiGd, and BTNO-1LiEr above 800 °C are basically maintained at the same level, that is, the degree of depolarization is similar. The relative d_{33} of BTNO-1LiLa is higher than that of other ceramics at the same temperature, which means that the degree of depolarization is small. This may be because the base structure formula of BTNO ceramics is $(Bi_2O_2)^{2+}(A_{n-1}B_nO_{3n+1})^{2-}$, and Bi^{3+} exists in both perovskite layer and bismuth layer. La^{3+} can form $(La_2O_2)^{2+}$ layer (La_2MoO_6) similar to bismuth layer [26], so it is regarded as entering Bi (4a) site of perovskite layer and Bi (8b) site of bismuth layer at the same time. Since

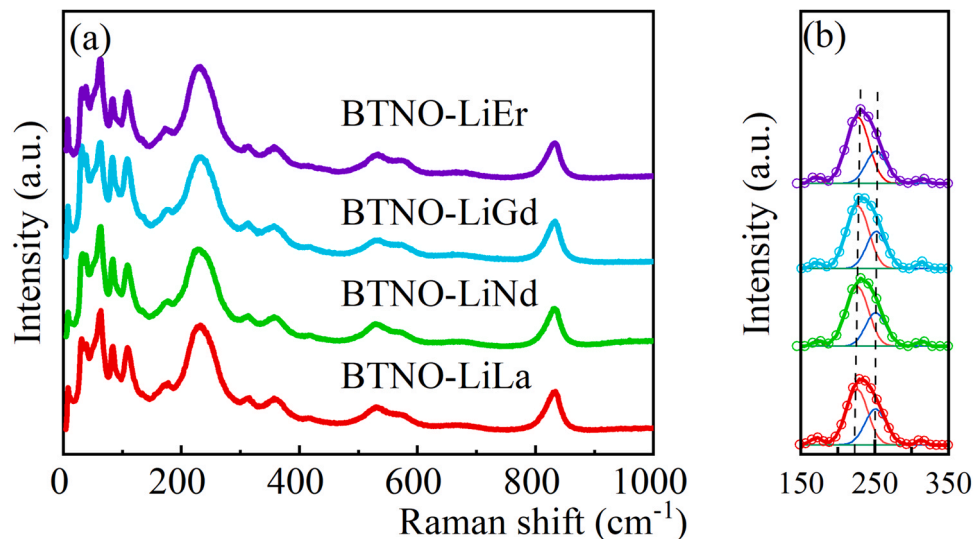


Fig. 3. Raman spectra of BTNO-1LiLn (Ln = La, Nd, Gd, and Er) ceramics at 25 °C in the range of 0 cm^{-1} ~ 1000 cm^{-1} and (b) the sum and peak positions of Lorentz fitting in the range of 150 cm^{-1} ~ 350 cm^{-1} .

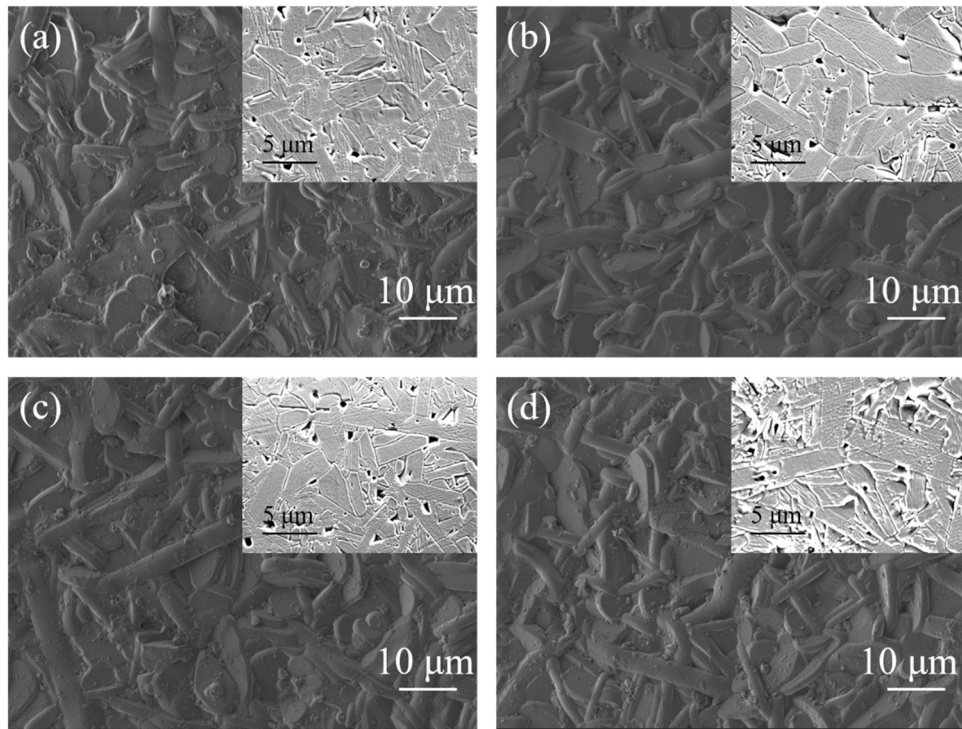


Fig. 4. SEM images of BTNO-1LiLn ceramics where (a) La (b) Nd, (c) Gd, and (d) Er. Inserted is the corresponding SEM image after thermal etching.

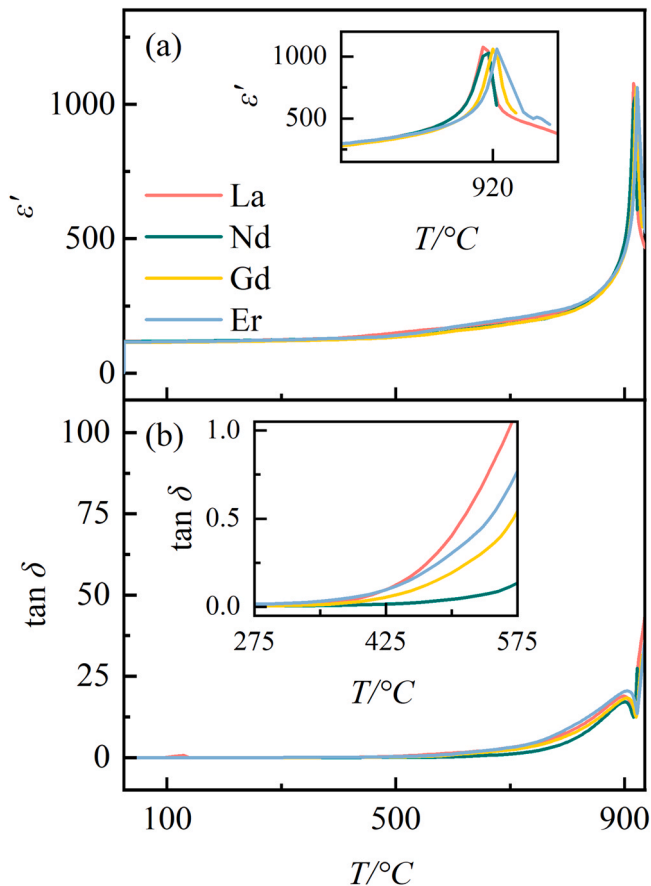


Fig. 5. (a) Dielectric permittivity (ϵ'), and (b) loss tangent ($\tan \delta$) for BTNO-1LiLn (Ln = La, Nd, Gd, and Er) ceramics from 25–935 °C under 100 kHz.

Nd³⁺, Gd³⁺, and Er³⁺ do not have 6s² lone pair electrons like Bi³⁺, they and Li⁺ are considered to only enter the Bi (4a) site of the perovskite layer. The difference of entry sites between La and other lanthanide elements may be the reason why BTNO-1LiLa has better anti depolarization performance.

As shown in Table 2, a series of properties of poled BTNO-1LiLn (Ln = La, Nd, Gd, and Er) ceramics can be calculated by the following formula. The electromechanical coupling factor (k) can reflect the coupling effect between mechanical energy and electrical energy in piezoelectric ceramics. The larger the value is, the better the conversion effect is. According to IEEE standard [27], the plane electrical coupling factor (k_p) and thickness electrical coupling factor (k_t) are calculated by resonance and anti-resonance method,

$$\frac{1}{k_p^2} = 0.395 \frac{f_r}{f_a - f_r} + 0.574 \quad (1)$$

$$k_t^2 = \frac{\pi f_r}{2 f_a} \tan \frac{\pi(f_a - f_r)}{2 f_a} \quad (2)$$

in the formula, f_a is the anti-resonant frequency, f_r is the resonant frequency, respectively.

Relative density (ρ) is an important performance index of reactive piezoelectric ceramics,

$$\rho = \rho_{the}/\rho_{exp} \times 100\% = \frac{\sum w_i}{a \times b \times c} \frac{m_0 \rho_e}{m_1 - m_2} \times 100\% \quad (3)$$

in this formula, the molecule is the theoretical density of ceramics (ρ_{the}), w_i is the weight of the atom at the i -th lattice position in a single cell. For the two-layer orthogonal bismuth layered structure compound with space group $A2_1am$, the i value is 56, the product of a , b , and c is the cell volume. The denominator is the experimental density of ceramics (ρ_{exp}), (ρ_e) is the density of ethanol, m_0 is the mass of the sample after drying at 80 °C for 2 h, m_1 is the mass of the sample in air after ethanol soaking, and m_2 is the mass of the sample

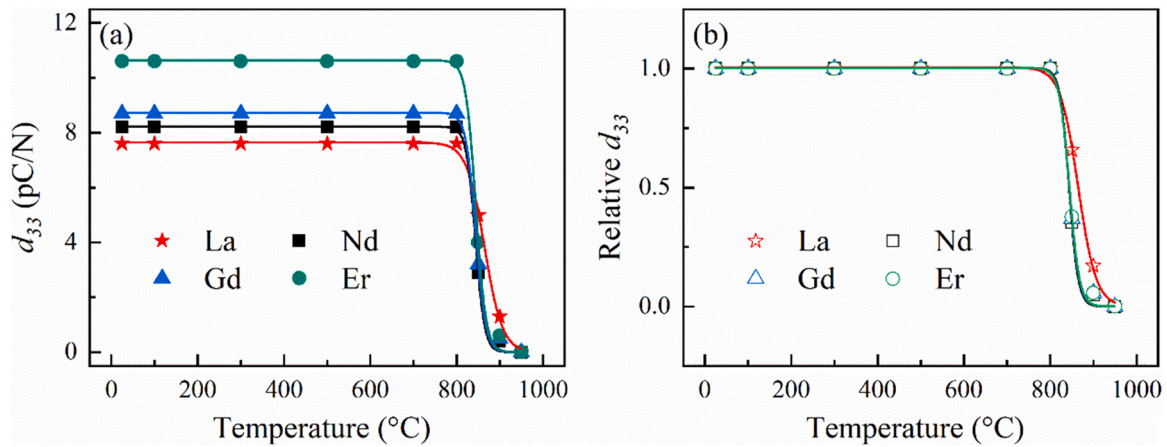


Fig. 6. The temperature dependence of piezoelectric coefficient (d_{33}) of poled BTNO-1LiLn ceramics, (a) the change of d_{33} , (b) the change of relative d_{33} .

Table 2

The curie temperature (T_c), dielectric permittivity (ϵ'), piezoelectric coefficient (d_{33}), plane electromechanical coupling factor (k_p), thickness electromechanical coupling factor (k_t), relative density (ρ), and tolerance factor (t) of poled BTNO-1LiLn (Ln = La, Nd, Gd, and Er) ceramics.

Content	T_c (°C)	ϵ'	d_{33} (pC/N)	k_p (%)	k_t (%)	ρ (%)	t
La	915.4	1077.04	7.6	5.38090	5.30783	93.61591	0.971671
Nd	918.2	1028.63	8.2	5.18861	5.11827	93.46743	0.971514
Gd	920.0	1062.64	8.7	5.18792	5.11758	93.36667	0.971433
Er	921.8	1062.94	10.6	5.18746	5.11714	93.44618	0.971355

in ethanol after ethanol soaking. The division of the two is the relative density.

Tolerance factor (t) can reflect the deviation degree between ceramic and ideal structure [28],

$$t = \frac{R_A + R_O}{\sqrt{2(R_B + R_O)}} \quad (4)$$

in this formula, R_A , R_B , and R_O represent the Shannon ion radius of A-site cation, B-site cation and oxygen ion, respectively [20]. The t of BLSFs ranges from 0.87 to 0.99 [29].

According to the experience summary, there is an inverse proportional relationship between T_c and t of ceramics. With the increase of lanthanide atomic number, the smaller the ion radii under the same coordination number, so that t decreases with the decrease of composite ion radii, and T_c increases. The complex domain dynamics of BTNO-1LiLa is different from other ceramics due to the special occupation of La, which is the reason why its k_p and k_t are better than other ceramics [30].

Table 3

Rietveld refinement conditions and crystallographic data of BTNO-1LiLa and BTNO-1LiEr.

Compound	$\text{Bi}_{2.99}(\text{Li}_{0.5}\text{La}_{0.5})_{0.01}\text{TiNbO}_9$	$\text{Bi}_{2.99}(\text{Li}_{0.5}\text{Er}_{0.5})_{0.01}\text{TiNbO}_9$
Crystal system	Orthorhombic	Orthorhombic
a (Å)	5.409441	5.409946
b (Å)	5.453435	5.454065
c (Å)	25.164959	25.167761
V (Å ³)	742.3	742.6
Temperature (K)	298	298
Space group	$A2_1am$	$A2_1am$
No. of formula units per unit cell, Z	4	4
No. of data points	5134	5134
Reduced χ^2	7.615	7.718
R_{wp} (%)	8.22	8.63
R_p (%)	5.23	5.94

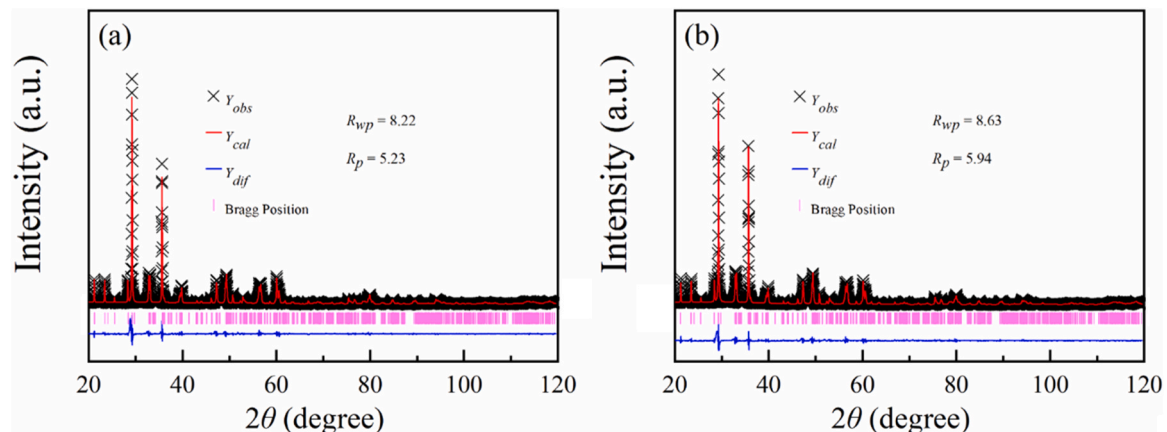


Fig. 7. The Rietveld refinement of (a) BTNO-1LiLa and (b) BTNO-1LiEr.

Table 4

Atomic coordinates, atomic occupancies and isotropic thermal parameters obtained by Rietveld refinement of BTNO-1LiLa at room temperature.

Atom	site	x	y	z	100U _{iso} /Å ²	Occupancy
Bi1	4a	0.005191	0.760697	0.500000	1.535	0.9916
Li1	4a	0.005191	0.760697	0.500000	1.535	0.0050
La1	4a	0.005191	0.760697	0.500000	1.535	0.0034
Bi2	8b	0.019165	0.734027	0.366367	0.958	0.9984
La2	8b	0.019165	0.734027	0.366367	0.958	0.0016
Nb1	8b	0.057321	0.277048	0.588122	1.252	0.5
Ti1	8b	0.057321	0.277048	0.588122	1.252	0.5
O1	4a	0.573420	0.221793	0.500000	2.951	1
O2	8b	-0.052984	0.382209	0.661779	2.951	1
O3	8b	0.333439	0.477627	0.240866	2.951	1
O4	8b	0.310011	0.587545	0.567523	2.951	1
O5	8b	0.318078	0.064432	0.584473	2.951	1

Table 5

Atomic coordinates, atomic occupancies and isotropic thermal parameters obtained by Rietveld refinement of BTNO-1LiEr at room temperature.

Atom	site	x	y	z	100U _{iso} /Å ²	Occupancy
Bi1	4a	-0.003487	0.761264	0.500000	2.376	0.99
Li1	4a	-0.003487	0.761264	0.500000	2.376	0.005
Er1	4a	-0.003487	0.761264	0.500000	2.376	0.005
Bi2	8b	0.002048	0.739823	0.699661	1.478	1
Nb1	8b	0.046204	0.255923	0.589000	3.306	0.5
Ti1	8b	0.046204	0.255923	0.589000	3.306	0.5
O1	4a	-0.037635	0.503787	0.500000	2.142	1
O2	8b	0.054913	0.209193	0.663665	2.142	1
O3	8b	0.185921	0.337805	0.243067	2.142	1
O4	8b	0.263145	0.544666	0.573055	2.142	1
O5	8b	0.356200	-0.088456	0.579364	2.142	1

Table 6

The bond length (Å) obtained by Rietveld refinement, and the valence state (V_i) of each ion at the lattice position by calculation for BTNO-1LiLa and BTNO-1LiEr.

Bond Type		Bond lengths (Å)		Bond valence (V _i)	
		La	Er	La	Er
A1-O	A1-O1	3.29113(0) × 1	2.91685(5) × 1		
		3.45700(7) × 1	2.41914(4) × 1		
		2.56631(7) × 1	2.53569(5) × 1		
		2.74404(1) × 1	2.87877(6) × 1		
	A1-O4	2.74889(4) × 2	2.62640(2) × 1		
		2.60335(3) × 2	2.78303(3) × 2	Bi1, 2.596	Bi1, 2.265
A2-O	A1-O5	2.54000(9) × 2	2.91263(3) × 2	Li1, 0.476	Li1, 0.415
		2.51819(2) × 2	2.77837(3) × 2	La1, 3.209	Er1, 1.953
	A2-O2	2.16525(5) × 1	2.59798(5) × 1		
		2.62787(5) × 1	2.73057(6) × 1		
	A2-O3	2.11338(4) × 1	1.87115(2) × 1		
		2.12071(2) × 1	2.27636(3) × 1		
B-O	A2-O3	2.43552(4) × 1	2.80621(5) × 1	Bi2, 3.912	Bi2, 3.304
		2.70063(4) × 1	2.55685(2) × 1	La2, 4.837	
	B-O1	2.03653(5) × 1	2.65529(2) × 1		
		2.02978(2) × 1	1.89694(2) × 1		
		1.60562(3) × 1	1.92065(3) × 1		
		2.23123(4) × 1	2.00463(4) × 1		
	B-O5	1.82816(3) × 1	1.59631(2) × 1	Ti, 4.452	Ti, 4.421
	B-O5	2.26953(5) × 1	2.52967(4) × 1	Nb, 5.771	Nb, 5.452

In order to further analyze the principle of improving the performance of BTNO, BTNO-1LiLa with special occupation and BTNO-1LiEr with best piezoelectric properties were refined by XRD with GSAS-EXPGUI [31,32]. As can be seen from Fig. 7, the final refined contour well matches the experimental XRD data.

Table 3 lists the crystallographic data, reliability factor R value and refinement conditions of refined BTNO-1LiLa and BTNO-1LiEr. The R_{wp} , R_p , and reduced χ^2 values of BTNO-1LiLa are 7.615, 8.22, and 5.23, respectively. The R_{wp} , R_p , and reduced χ^2 values of BTNO-1LiEr

are 7.718, 8.63, and 5.94, respectively. Table 4 and Table 5 list the atomic coordinates, atomic occupancies and isotropic thermal parameters of BTNO-1LiLa and BTNO-1LiEr after Rietveld refinement. Table 6 shows the bond length (Å) of BTNO-1LiLa and BTNO-1LiEr, and the valence states (V_i) of positive ions at each position are calculated by using Brown & Altermatt equation [33]:

$$V_i = \sum s_{ij} = \sum \exp[(r_0 - r_{ij})/B] \quad (5)$$

among them, the empirical constant B is generally 0.37, and r_{ij} is the bond length observed experimentally, while the bond constant r_0 of each ion A-X type can be referred to the work of Brown & Altermatt [33]. It can be seen that the V_i of ions in the bismuth layer and B-site increase and form overbonding, and the V_i of Bi and Li ions at A-site are reduced to form underbonding. But the V_i of La and Er are opposite, the valence changes of BTNO-1LiEr are similar to those of introducing (Li_{0.5}Sm_{0.5})²⁺ into BTNO. The reason for this abnormal situation is that after La³⁺ replaces Bi³⁺, the 2p orbital hybridization between La³⁺ and O²⁻ is weak, which will indirectly affect the hybridization between B-site ion and O²⁻ ion 2p orbital, and enhance the short-range repulsion between ions [34,35]. And under the same coordination number, the ion radius of La³⁺ is close to that of Bi³⁺, resulting in the bond length formed by the A-site ion of BTNO-1LiLa and O in BO6 octahedron will be significantly different from that of BTNO-1LiEr. At the same time, the r_0 of La-O is the longest, which is reflected in the increase of V_i.

According to Tables 3, 4, 5, and 6, the contribution of each ion in BTNO-1LiLa and BTNO-1LiEr to spontaneous polarization (P_s) are calculated based on Shimakawa formula [34,36–38],

$$P_s = \sum_i \frac{m_i \times \Delta x_i \times Q_i e}{V} \quad (6)$$

in the formula, m_i represents the number of equivalent positions in the unit cell, Δx_i is the displacement of the corresponding position of the atom relative to the parent phase along the a-axis, $Q_i e$ is the electricity valence of the i-th component ion, and V is the volume of a unit cell.

As shown in Fig. 8, O1 and O4 in the perovskite layer of BTNO-1LiEr provide a total of 39.7% of all P_s, which is consistent with the fact that Withers et al. found that the generation of underbonding of A-site ions in the prototype structure will cause the displacement of BO6 octahedron in the perovskite layer along the a-axis relative to A-site ions, which can provide large P_s [39]. The anisotropic movement of Bi2 and O3 in the bismuth layer provide 20.4 % and 37.2 % of the total P_s, respectively. Because Bi2 in the bismuth layer will also bond with O2 in the perovskite layer. When the BO6 octahedron moves integrally, O2 in it will also drive Bi2 in the bismuth layer to move, so that it can also provide a certain amount of P_s. The contribution mode of atomic displacement and spontaneous polarization at each position of BTNO-1LiEr is basically similar to that of introducing Sm into BTNO, while the mode of BTNO-1LiLa is significantly different. This is because the introduction of La enhances the short-range repulsion between ions, increases the symmetry of BO6 octahedron, increases the deflection of polarization along the c-axis and decreases the polarization along the a-axis. However, due to the structural characteristics of BLSFs, the atomic displacement along the b-axis and c-axis will not produce ferroelectric polarization or the ferroelectric polarization will be very small due to the existence of slip and mirror plane [37]. Moreover, La³⁺ enters the bismuth layer, so that O3 located in the bismuth layer deteriorates 47.4 % of all P_s, resulting in its smaller d₃₃. The O2 related to La2 in bismuth layer in perovskite layer provides 67.8 % of all P_s, this part of P_s comes from the joint action of Bi1/Li1/La1 in perovskite layer and Bi2/La2 in bismuth layer. It is not only from the distortion of BO6 octahedron, so it has good anti depolarization ability, which is mutually demonstrated with Fig. 6.

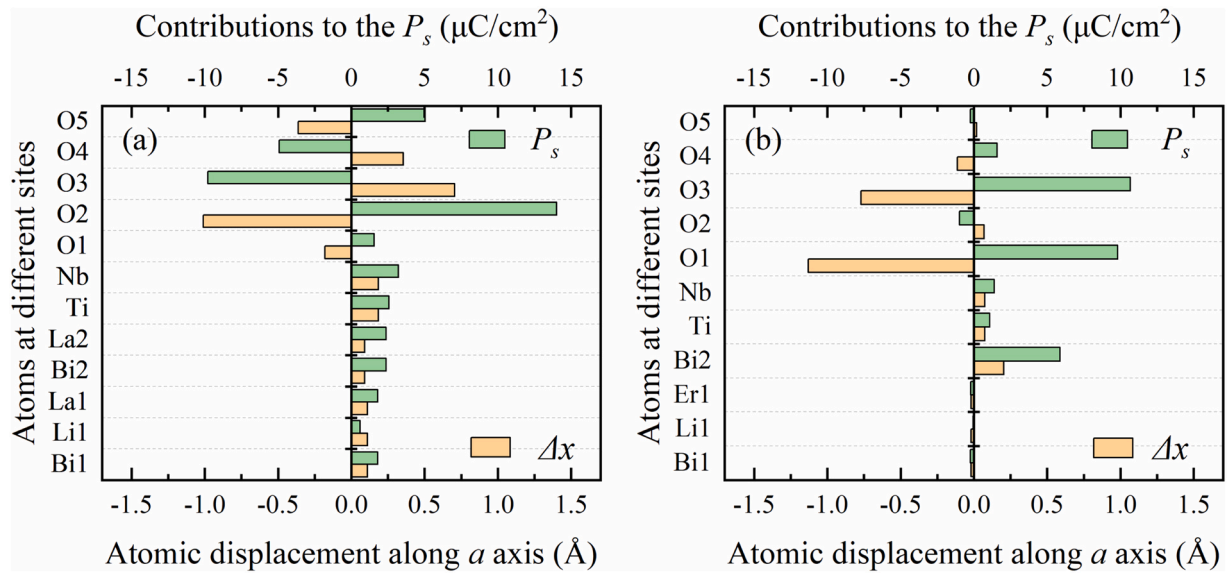


Fig. 8. The displacement of ions along the a -axis and contributions to spontaneous polarization (P_s) in (a) BTNO-1LiLa and (b) BTNO-1LiEr.

The fatigue characteristics of BTNO-1LiLa and BTNO-1LiEr ceramics under ferroelectric polarization 10^7 at 40 °C and 120 °C are shown in Fig. 9. The fatigue electric field of 100 kV/cm is used to make the ferroelectric domain reverse enough, and the fatigue frequency of 100 Hz makes the reversal of ferroelectric domain keep up with the change of electric field. For BTNO-1LiLa, it showed very

good cycle stability, its normalized P_s decreased by 4 % at 40 °C and 11 % at 120 °C, respectively. With the increase of fatigue cycle times, normalized P_s of BTNO-1LiEr decreases sharply, which decreases by 11 % at 40 °C and 39 % at 120 °C. In our previous work, we found that undoped BTNO decreased by 12 % at 40 °C and 14 % at 120 °C. The introduction of La will significantly improves the fatigue stability of

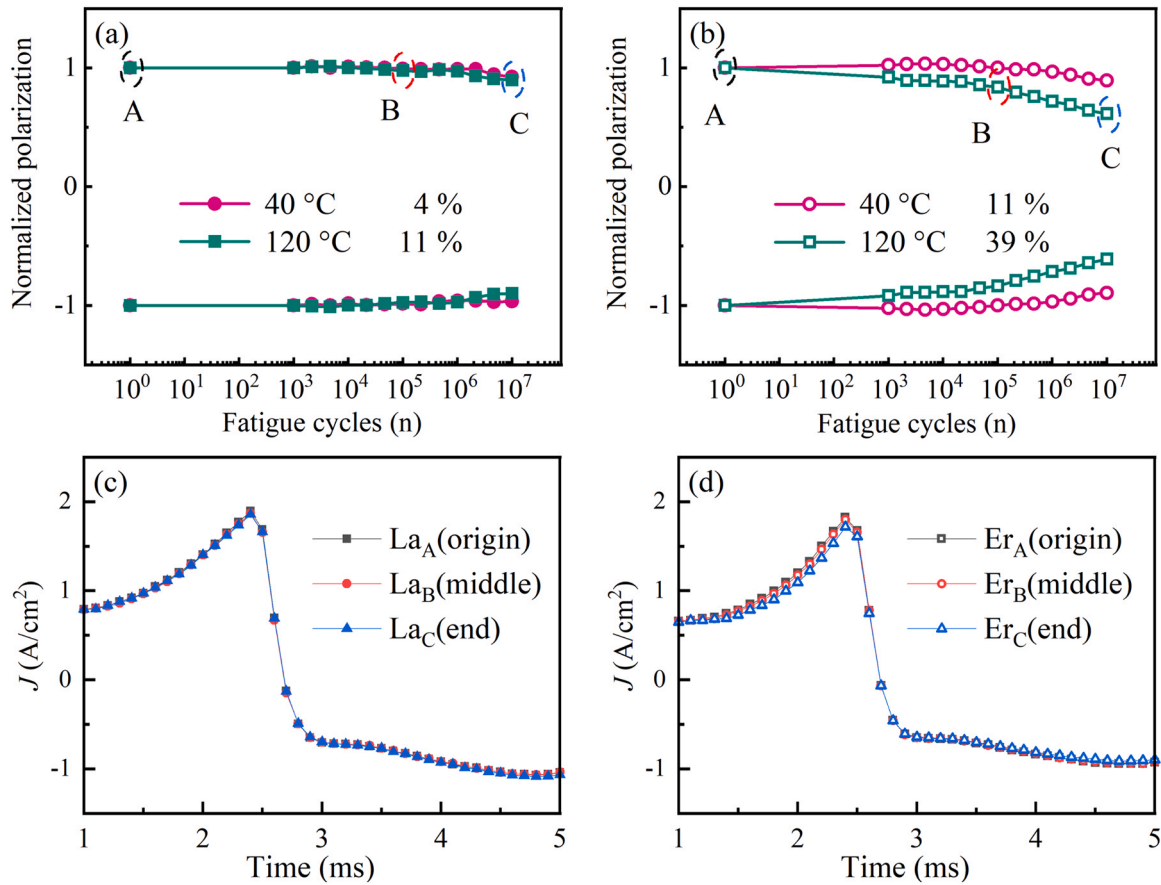


Fig. 9. The displacement of ions along the a -axis and contributions to spontaneous polarization (P_s) in (a) BTNO-1LiLa and (b) BTNO-1LiEr.

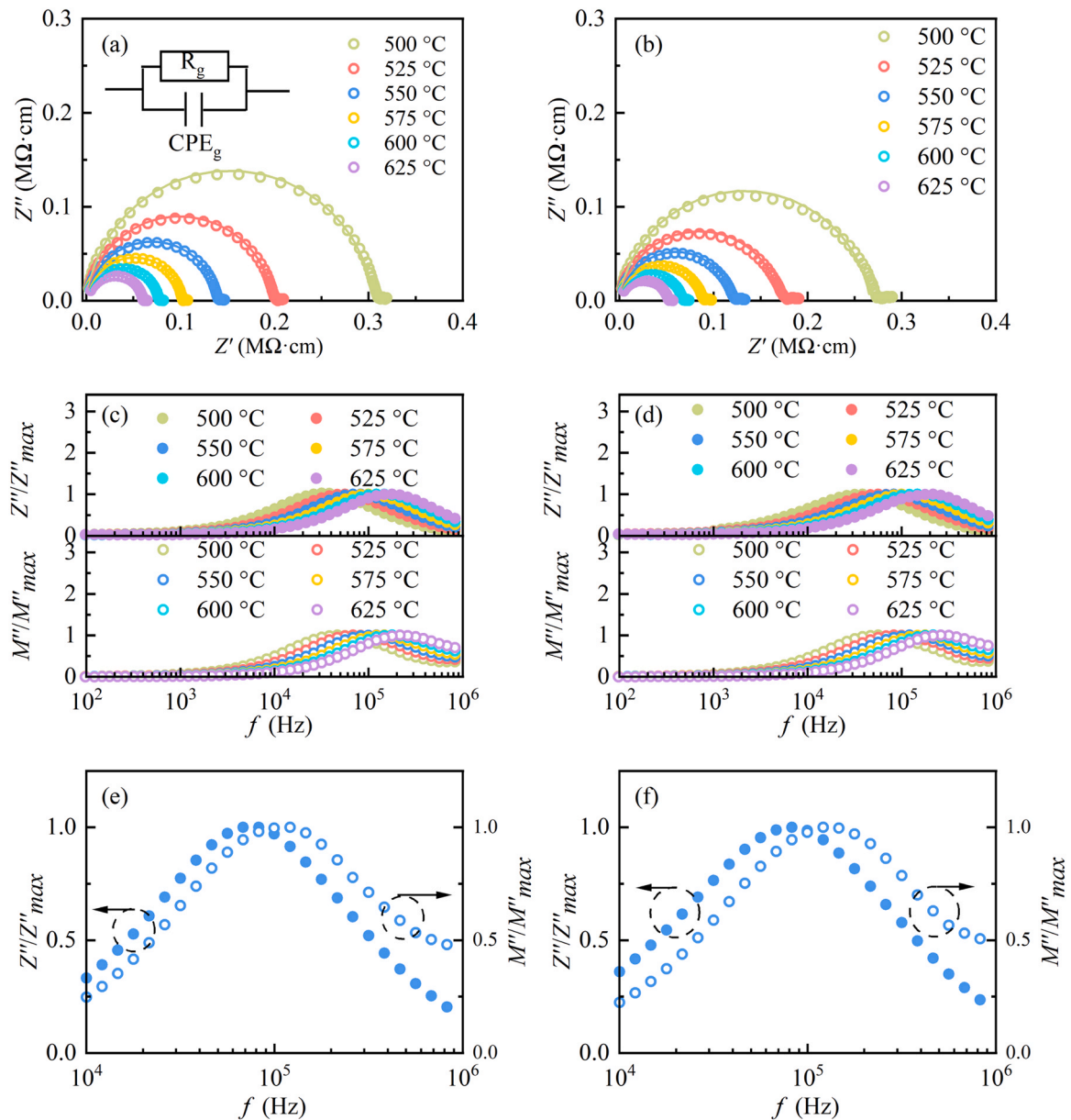


Fig. 10. AC impedance spectra of (a) BTNO-1LiLa and (b) BTNO-1LiEr ceramics at the temperature of 500 °C ~ 625 °C (the symbols: experimental data, the solid lines: fitting lines). Z''/Z''_{max} and M''/M''_{max} variation with frequency of (c) BTNO-1LiLa and (d) BTNO-1LiEr at the temperature of 500 °C ~ 625 °C. Comparison of Z''/Z''_{max} and M''/M''_{max} of (e) BTNO-1LiLa and (f) BTNO-1LiEr at the temperature 550 °C.

BTNO, while the occupation of Er in BTNO ceramics is the same as that of Sm, so it shows similar fatigue characteristics. At present, it is considered that the polarization fatigue in ferroelectric materials is caused by domain wall pinning induced by point defects [40]. In order to illustrate the difference of ferroelectric domain inversion of ceramics during fatigue, we selected three states of fatigue origin, middle, and end, corresponding to points A, B, and C in Fig. 9(a) and (b), respectively. The polarization current density (J) of the initial domain inversion of BTNO-1LiLa and BTNO-1LiEr during fatigue at 120 °C changes with time, as shown in Fig. 9(c) and (d). It can be seen that with the increase of the number of fatigue cycles, the J_{max} value of BTNO-1LiEr decreases significantly, while that of BTNO-1LiLa does not change significantly. This shows that there is defect agglomeration pinning the domain wall of BTNO-1LiEr ceramics during fatigue cycle, resulting in the reduction of the number of reversible ferroelectric domains [41–43]. The fatigue characteristics of BTNO-1LiLa and BTNO-1LiEr are mutually demonstrated with the changes of relative d_{33} in Fig. 6(b) and the calculation results of P_s in Fig. 8.

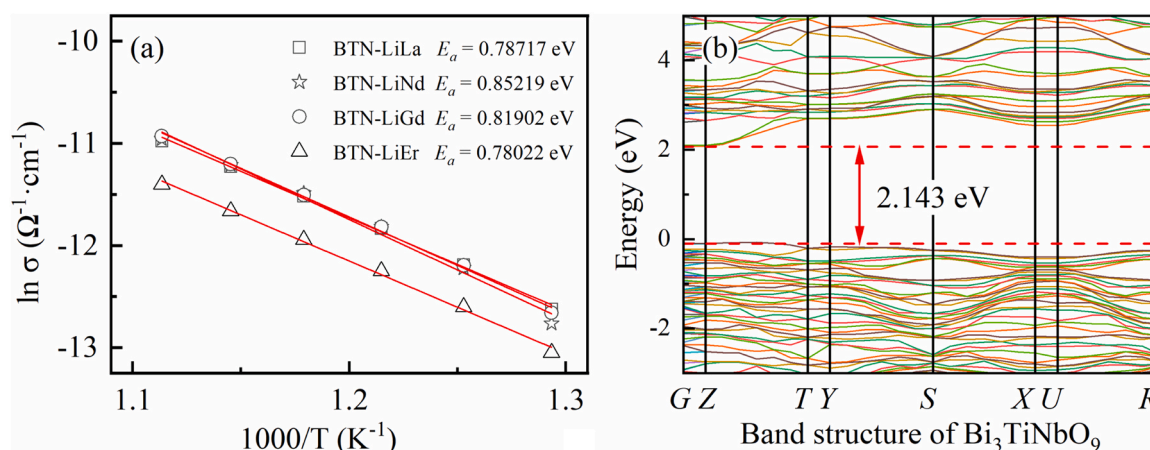
3.3. Impedance analysis

Generally speaking, the grains and grain boundaries of polycrystalline ceramics have different conductive mechanisms [44]. In order to further analyze the different AC impedance characteristics of btno-1liLa and btno-1liEr, the AC impedance spectra at 500 °C ~ 625 °C were measured, and the impedance data obtained from the experiment were fitted based on the circuit inserted in Fig. 10 (a), as shown in Table 7. It can be observed from Fig. 10 that there is no significant difference between the two groups of images, which shows that the doping elements entering the bismuth layer or perovskite layer will not significantly affect the AC impedance characteristics of ceramics. Fig. 10 (a) and (b) show the composite impedance spectra of BTNO-1LiLa and BTNO-1LiEr, both of which are single semicircles reflecting electric homogeneity. Fig. 10 (c) and (d) show the variation curves of impedance imaginary part (Z'') and modulus imaginary part (M'') of BTNO-1LiLa and BTNO-1LiEr with frequency. The frequencies corresponding to the peaks of the two

Table 7

Impedance parameters calculated from the complex impedance plots for BTNO-1LiLn (Ln = La, Nd, Gd, and Er) in the temperature range of 500 ° ~ 625 °.

TEMP (°C)	La			Nd			Gd			Er		
	R _{obs} (Ω)	R _{cal} (Ω)	Error (%)	R _{obs} (Ω)	R _{cal} (Ω)	Error (%)	R _{obs} (Ω)	R _{cal} (Ω)	Error (%)	R _{obs} (Ω)	R _{cal} (Ω)	Error (%)
500	17,223	16,836	0.53	22,282	21,452	0.74	26,937	26,355	0.56	23,183	22,347	0.72
525	11,146	10,895	0.47	12,985	12,474	0.61	16,849	16,604	0.47	14,810	14,300	0.65
550	7769.8	7619	0.43	8655.7	8313	0.55	11,551	11,383	0.42	10,352	10,041	0.59
575	5634.3	5546	0.40	6226.7	5992	0.53	8477.1	8315	0.41	7591.9	7396	0.54
600	4243.8	4155	0.44	4676.3	4539	0.52	6242.7	6145	0.38	5742.9	5582	0.54
625	3296.4	3238	0.40	3619.6	3523	0.52	4759.6	4682	0.37	4436.4	4299	0.55

**Fig. 11.** (a) The Arrhenius-type plots of electrical conductivity of BTNO-1LiLn (Ln = La, Nd, Gd, and Er) ceramics (the symbols: experimental data, the solid lines: fitting lines). (b) Band structure of Bi₃TiNbO₉.

curves are very close, indicating that the electronic structure of the sample is relatively uniform [45]. Fig. 10 (e) and (f) show the variation curves of Z'' and M'' of BTNO-1LiLa and BTNO-1LiEr with frequency at 550 °C. There is a certain mismatch between the two peaks, which shows that the local conductance of short-range motion of charge carriers is the main mechanism [46].

In order to further analyze the difference of its conductivity mechanism, the activation energy (E_a) of Bi_{2.99}(Li_{0.5}Ln_{0.5})_{0.01}TiNbO₉ (BTNO-1LiLn, Ln = La, Nd, Gd, and Er) ceramics in the range of 500 °C ~ 625 °C are calculated according to Arrhenius formula:

$$\sigma = \sigma_0 \exp(-E_a/kT) \quad (7)$$

among them, σ represents conductivity, σ_0 is the pre-exponential constant, k and T are the Boltzmann constant and the corresponding temperature, respectively.

As can be seen from Fig. 11 (a), the calculated E_a is 0.7 eV ~ 0.9 eV. Then taking the crystal structure of Bi₃TiNbO₉ reported by JG Thompson et al. [47], the electronic energy band of Bi₃TiNbO₉ is calculated after geometric optimization by Materials Studio. As shown in Fig. 11 (b), the valence band top and conduction band bottom of BTNO ceramics are at point Z, which belongs to direct band gap semiconductor, and the band gap (E_g) is about 2.143 eV. The E_a of BTNO doped with (Li_{0.5}Ln_{0.5})²⁺ ions are slightly less than half of its E_g . This is because Bi with three valence electrons at A-site in BTNO is replaced by an atomic group with two valence electrons (Li_{0.5}Ln_{0.5}), but it is still bonded with O at three different positions, and an electron vacancy will appear in one of the bonds. This vacancy can move freely in the crystal by capturing electrons in other bonds and become an electron hole in the valence band. At the same time, the (Li_{0.5}Ln_{0.5}) atomic group accepts an electron and becomes a monovalent negative ion, forming a fixed negative charge center, providing a localized electronic state and forming the acceptor level [48]. (Li_{0.5}Ln_{0.5})²⁺ ions play the role of acceptor impurities in the A-

site substitution of BTNO ceramics. Therefore, the acceptor impurities can ionize holes in the valence band without too high temperature. The acceptor level makes BTNO-1LiLn ceramics need lower E_a to realize electrical conductivity.

4. Conclusion

Bismuth layer structured ferroelectrics (BLSFs) Bi_{2.99}(Li_{0.5}Ln_{0.5})_{0.01}TiNbO₉ (BTNO-1LiLn, Ln = La, Nd, Gd, and Er) high temperature piezoelectric ceramics were synthesized by solid state reaction sintering, and their microstructure and electrical properties were studied in detail. It is found that BTNO-1LiLa has excellent anti depolarization performance due to the difference in the occupation of La³⁺ and other Ln³⁺ (Nd³⁺, Gd³⁺, and Er³⁺) in BTNO. However, the special occupation of La³⁺ makes O3 in bismuth layer seriously deteriorate the spontaneous polarization (P_s), and the larger ion radius of La³⁺ makes the overall distortion of BO₆ octahedron along the a -axis smaller, resulting in the piezoelectric coefficient (d_{33}) of BTNO-1LiLa is only 7.6 pC/N, while that of BTNO-1LiEr reached 10.6 pC/N. Finally, based on the first principle, the band structure of BTNO pure ceramic is calculated by Materials Studio, and the band gap (E_g) is 2.143 eV, while the activation energy (E_a) of BTNO-1LiLn is between 0.7 eV ~ 0.9 eV, which shows that the introduction of (Li_{0.5}Ln_{0.5})²⁺ in BTNO will transform it into p -type semiconductor, and the conductivity is realized by the movement of electron holes. This work provides theoretical support for the design of high temperature piezoelectric ceramics with better performance in the future.

CRediT authorship contribution statement

The author confirms the manuscript has been approved by all authors for publication and there is no conflict of interest exists in the submission of this manuscript. I would like to declare on behalf of

my co-authors that the work described was original research that has not been published previously, and not under consideration for publication elsewhere, in whole or in part. All the authors listed have approved the manuscript that is enclosed.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Key Research and Development Project (2020YFC1521900 and 2020YFC1521904), the Shaanxi Provincial Science Foundation (2021GXLH-01-11), the National Nature Science Foundation (51902258 and 51902259), the SKLSP Project (2019-TZ-04), and the 111 Program (B08040) of MOE of China. We would also like to thank the Analytical & Testing.

References

- [1] X. Jiang, K. Kim, S. Zhang, J. Johnson, G. Salazar, High-temperature piezoelectric sensing, *Sensors* 14 (2013) 144–169.
- [2] S. Zhang, F. Yu, Piezoelectric materials for high temperature sensors, *J. Am. Ceram. Soc.* 94 (2011) 3153–3170.
- [3] B. Park, B. Kang, S. Bu, T. Noh, J. Lee, W. Jo, Lanthanum-substituted bismuth titanate for use in non-volatile memories, *Nature* 401 (1999) 682–684.
- [4] P. Fang, H. Fan, J. Li, F. Liang, Lanthanum induced larger polarization and dielectric relaxation in Aurivillius phase $\text{SrBi}_{2-x}\text{La}_x\text{Nb}_2\text{O}_{9-x}$ ferroelectric ceramics, *J. Appl. Phys.* 107 (2010) 064104.
- [5] R. Turner, P.A. Fuierer, R. Newnham, T.R. Shrout, Materials for high temperature acoustic and vibration sensors: a review, *Appl. Acoust.* 41 (1994) 299–324.
- [6] D. Damjanovic, Materials for high temperature piezoelectric transducers, *Curr. Opin. Solid State Mater. Sci.* 3 (1998) 469–473.
- [7] S. Zhang, F. Li, F. Yu, Piezoelectric materials for cryogenic and high-temperature applications, *Structural Health Monitoring (SHM) in Aerospace Structures*, Elsevier, 2016, pp. 59–93.
- [8] Uchino, Kenji, *Piezoelectric Actuators and Ultrasonic Motors*, Piezoelectric Actuators and Ultrasonic Motors, 2011.
- [9] B. Aurivillius, Mixed Bismuth Oxides with Layer Lattices I. The Structure Type of $\text{CaNb}_2\text{Bi}_{20}\text{O}_9$, *Ark. kemi* 1 (1949) 463–480.
- [10] Y. Wu, M.J. Forbess, S. Seraji, S.J. Limmer, T.P. Chou, C. Nguyen, G. Cao, Doping effect in layer structured $\text{SrBi}_2\text{Nb}_2\text{O}_9$ ferroelectrics, *J. Appl. Phys.* 90 (2001) 5296–5302.
- [11] B. Aurivillius, P. Fang, Ferroelectricity in the compound $\text{Ba}_2\text{Bi}_4\text{Ti}_5\text{O}_{18}$, *Phys. Rev.* 126 (1962) 893.
- [12] X. Zeng, C.F. eng, Z. Peng, X. Xing, Crystal structure and electrical properties of (Li, Ce, Nd)-multidoped $\text{CaBi}_2\text{Nb}_2\text{O}_9$ high temperature ceramics, *Ceram. Int.* 44 (2017).
- [13] Z. Peng, Q. Chen, Y. Wang, D. Xin, D. Xian, J. Zhu, Enhancement of piezoelectric properties of (LiCePr)-multidoped $\text{CaBi}_2\text{Nb}_2\text{O}_9$ high temperature ceramics, *Mater. Lett.* 107 (2013) 14–16.
- [14] Z. Zhou, X. Dong, S. Huang, H. Yan, Dielectric relaxation of La^{3+} -Modified $\text{Bi}_3\text{TiNbO}_9$ aurivillius phase ceramics, *J. Am. Ceram. Soc.* 89 (2006) 2939–2942.
- [15] H. Yan, H. Zhang, R. Ubic, M.J. Reece, J. Liu, Z. Shen, Z. Zhang, A lead-free high-temperature ferroelectric ceramic, $\text{CaBi}_2\text{Nb}_2\text{O}_9$, *Adv. Mater.* 17 (2005) 1261–1265.
- [16] S. Swartz, W.A. Schulze, J.V. Biggers, Fabrication and electrical properties of grain oriented $\text{Bi}_4\text{Ti}_3\text{O}_{12}$ ceramics, *Ferroelectrics* 38 (1981) 765–768.
- [17] H. Chen, J. Zhai, Enhancing piezoelectric performance of $\text{CaBi}_2\text{Nb}_2\text{O}_9$ ceramics through microstructure control, *J. Electron. Mater.* 41 (2012) 2238–2242.
- [18] H. Chen, F. Fu, J. Zhai, Fabrication and piezoelectric property of highly textured $\text{CaBi}_2\text{Nb}_2\text{O}_9$ ceramics by tape casting, *Jpn. J. Appl. Phys.* 50 (2011) 050207.
- [19] D. Hou, H. Fan, A. Zhang, Y. Jia, W. Wang, Role of BO_6 octahedral distortion on high temperature piezoelectric properties in $\text{Bi}_3\text{-x}(\text{Li}_0.5\text{Sm}_0.5)\text{xTiNbO}_9$, *Ceram. Int.* (2022).
- [20] R.D.J. Shannon, Revised effective ionic radii and systematic study of inter atomic distances in halides and chalcogenides, *Acta Crystallogr. A* 32 (1976) 751–767.
- [21] Q. Wang, C.M. Wang, J.F. Wang, S. Zhang, High performance Aurivillius-type bismuth titanate niobate ($\text{Bi}_3\text{TiNbO}_9$) piezoelectric ceramics for high temperature applications, *Ceram. Int.* (2016) 6993–7000.
- [22] S. Mehraj, M.S. Ansari, Structural, electrical and magnetic properties of (Fe, Co) co-doped SnO_2 diluted magnetic semiconductor nanostructures, *Phys. E: Low-Dimens. Syst. Nanostruct.* 65 (2015) 84–92.
- [23] S. Mehraj, M.S. Ansari, A.A. Al-Ghamdi, Annealing dependent oxygen vacancies in SnO_2 nanoparticles: structural, electrical and their ferromagnetic behavior, *Mater. Chem. Phys.* 171 (2016) 109–118.
- [24] A. Azam, A.S. Ahmed, M.S. Ansari, A.H. Naqvi, Study of electrical properties of nickel doped SnO_2 ceramic nanoparticles, *J. Alloy. Compd.* 506 (2010) 237–242.
- [25] S. Mehraj, M.S. Ansari, Rutile-type Co doped SnO_2 diluted magnetic semiconductor nanoparticles: structural, dielectric and ferromagnetic behavior, *Phys. B: Condens. Matter* 430 (2013) 106–113.
- [26] J. Mercurio, A. Souirti, M. Manier, B. Frit, Phase transitions and dielectric properties in some compounds with bismuth oxide layer structure, *Mater. Res. Bull.* 27 (1992) 123–128.
- [27] A.H. Meitzler, D. Berlincourt, F.S. Welsh, H.F. Tiersten, G.A. Coquin, A.W. Warner, IEEE Standard on Piezoelectricity, IEEE, (1988).
- [28] V.M. Goldschmidt, Geochemische Verteilungsgesetze der Elemente. VII. Die Gesetze der Kristallchemie, *Skrifter Norske vidensk. Akad. Oslo*.
- [29] I. Ismailzade, V. Nesterenko, F. Mirishli, P. Rustamov, X-ray Diffraction Study of the $\text{Bi}_4\text{Ti}_3\text{O}_{12}$ - BiFeO_3 System, *Kristallografiya* 12 (1967) 468–473.
- [30] K. Uchino, High electromechanical coupling piezoelectrics: relaxor and normal ferroelectric solid solutions, *Solid State Ion.* 108 (1998) 43–52.
- [31] A.C. Larson, R.B. Von Dreele, General structure analysis system (GSAS)(Report LAUR 86-748), Los Alamos, New Mexico: Los Alamos National Laboratory, (2004).
- [32] B.H. Toby, EXPGUI, a graphical user interface for GSAS, *J. Appl. Crystallogr.* 34 (2001) 210–213.
- [33] I. Brown, D. Altermatt, Bond-valence parameters obtained from a systematic analysis of the inorganic crystal structure database, *Acta Crystallogr. Sect. B: Struct. Sci.* 41 (1985) 244–247.
- [34] Y. Shimakawa, Y. Kubo, Y. Tauchi, H. Asano, Z. Hiroi, Crystal and electronic structures of $\text{Bi}_4\text{LaTi}_3\text{O}_{12}$ ferroelectric materials, *Appl. Phys. Lett.* 79 (2001) 2791–2793.
- [35] C.Y. Yau, R. Palan, K. Tran, R.C. Buchanan, Mechanism of polarization enhancement in La-doped $\text{Bi}_4\text{Ti}_3\text{O}_{12}$ films, *Appl. Phys. Lett.* 86 (2005) 658.
- [36] Y. Shimakawa, Y. Kubo, Y. Nakagawa, T. Kamiyama, H. Asano, F. Izumi, Crystal structures and ferroelectric properties of $\text{SrBi}_2\text{Ta}_2\text{O}_9$ and $\text{Sr}_0.8\text{Bi}_{1.2}\text{Ta}_2\text{O}_9$, *Appl. Phys. Lett.* 74 (1999) 1904–1906.
- [37] Y. Shimakawa, Y. Kubo, Y. Nakagawa, S. Goto, T. Kamiyama, H. Asano, F. Izumi, Crystal structure and ferroelectric properties of $\text{A Bi}_2\text{Ta}_2\text{O}_9$ (A = Ca, Sr, and Ba), *Phys. Rev. B* 61 (2000) 6559.
- [38] Y. Shimakawa, Y. Kubo, Y. Tauchi, T. Kamiyama, H. Asano, F. Izumi, Structural distortion and ferroelectric properties of $\text{SrBi}_2(\text{Ta}_{1-x}\text{Nb}_x)_2\text{O}_9$, *Appl. Phys. Lett.* 77 (2000) 2749–2751.
- [39] R. Withers, J. Thompson, A. Rae, The crystal chemistry underlying ferroelectricity in $\text{Bi}_4\text{Ti}_3\text{O}_{12}$, $\text{Bi}_3\text{TiNbO}_9$, and Bi_2WO_6 , *J. Solid State Chem.* 94 (1991) 404–417.
- [40] C. Doru, Ute Lupascu, Rabe, cyclic cluster growth in ferroelectric perovskites, *Phys. Rev. Lett.* 89 (2002) 187601–187601.
- [41] D. Damjanovic, M. Demartin, H. Shulman, M. Testorf, N. Setter, Instabilities in the piezoelectric properties of ferroelectric ceramics, *Sens. Actuators A: Phys.* 53 (1996) 353–360.
- [42] Q. Zhang, W. Pan, S. Jang, L. Cross, Domain wall excitations and their contributions to the weak-signal response of doped lead zirconate titanate ceramics, *J. Appl. Phys.* 64 (1988) 6445–6451.
- [43] Q. Zhang, H. Wang, N. Kim, L. Cross, Direct evaluation of domain-wall and intrinsic contributions to the dielectric and piezoelectric response and their temperature dependence on lead zirconate-titanate ceramics, *J. Appl. Phys.* 75 (1994) 454–459.
- [44] G. Yao, X. Wang, Y. Wu, L. Li, Nb-Doped 0.9 BaTiO_3 –0.1 (Bi 0.5 Na 0.5) TiO_3 Ceramics with Stable Dielectric Properties at High Temperature, *J. Am. Ceram. Soc.* 95 (2012) 614–618.
- [45] N. Maso, A.R. West, Electrical properties of Ca-doped BiFeO_3 ceramics: from p-type semiconduction to oxide-ion conduction, *Chem. Mater.* 24 (2012) 2127–2132.
- [46] D.K. Pradhan, R. Choudhary, C. Rinaldi, R. Katiyar, Effect of Mn substitution on electrical and magnetic properties of $\text{Bi}_{0.9}\text{La}_{0.1}\text{FeO}_3$, *J. Appl. Phys.* 106 (2009) 024102.
- [47] J. Thompson, A. Rae, R. Withers, D. Craig, Revised structure of $\text{Bi}_3\text{TiNbO}_9$, *Acta Crystallogr. Sect. B: Struct. Sci.* 47 (1991) 174–180.
- [48] L. Wu, A. Podpirka, J.E. Spanier, P.K. Davies, Ferroelectric, optical, and photovoltaic properties of morphotropic phase boundary compositions in the PbTiO_3 - BiFeO_3 - $\text{Bi}(\text{Ni}_{1/2}\text{Ti}_{1/2})\text{O}_3$ system, *Chem. Mater.* 31 (2019) 4184–4194.